Problem 1 | Problems T3_Diffusion.pdf #1 | Fick's 2nd Law · erf(z)

Cementation of Grade 1018 Steel Gear

Original statement · Problems T3_Diffusion.pdf

1. In a steel gear of grade 1018 (0.18 wt% C*) a cementation process is carried out at* 927°C*. Calculate the time required in order to increase the* C *content up to* 0.3 wt% *at* 0.6 mm *below the surface of the gear. Assume that the* C *content in the surface is* 1 wt%*. Data:* $D_{\text{C in } \gamma\text{-Fe}} = 1.28 \times 10^{-11}$ m^2/s. **Official Solution: t = 6636 s (1.84 h)**

PHASE 1 · HYPOTHESES & PARAMETERS

- **Semi-Infinite Medium:** The penetration depth $x = 0.6$ mm is small compared with the size of the part, allowing one-dimensional semi-infinite modeling [Session 5 Slide 20].
- **Constant Surface Concentration:** The carburizing atmosphere maintains $C(0,t) = C_s = 1.0$ wt% for $t > 0$.
- **Homogeneous Initial State:** The AISI 1018 steel core has initial carbon concentration $C(x,0) = C_0 = 0.18$ wt%.
- **Parameters:** $C_0 = 0.18$ wt%, $C_s = 1.00$ wt%, $C_x = 0.30$ wt%, $x = 0.6$ mm $= 6.0 \times 10^{-4}$ m, $D = 1.28 \times 10^{-11}$ m^2/s at $T = 927^\circ$C.

PHASE 2 · FORMULATION & PEDAGOGICAL STRATEGY

In non-steady-state diffusion into a semi-infinite solid with constant surface composition, Fick's Second Law yields [Session 5 Slide 20]:

$$\frac{C_s - C_x}{C_s - C_0} = \operatorname{erf}\left(\frac{x}{2\sqrt{Dt}}\right) = \operatorname{erf}(z)$$

We evaluate the concentration ratio, locate the corresponding argument z in the official error function table via exact linear interpolation, and solve for time t:

$$z = \frac{x}{2\sqrt{Dt}} \implies t = \frac{x^2}{4 \cdot z^2 \cdot D}$$

PHASE 3 · STEP-BY-STEP ANALYTICAL DERIVATION

1. Concentration Ratio:

$$\frac{C_s - C_x}{C_s - C_0} = \frac{1.00 - 0.30}{1.00 - 0.18} = \frac{0.70}{0.82} \approx 0.8536585 \implies \operatorname{erf}(z) = 0.8536585$$

2. Linear Interpolation in Error Function Table [Slide 21]:

- For $z_1 = 1.00 \implies \operatorname{erf}(z_1) = 0.8427$
- For $z_2 = 1.10 \implies \operatorname{erf}(z_2) = 0.8802$

$$\frac{z - 1.00}{1.10 - 1.00} = \frac{0.8536585 - 0.8427}{0.8802 - 0.8427} \implies \frac{z - 1.00}{0.10} = \frac{0.0109585}{0.0375} \approx 0.292227$$

$$z = 1.00 + 0.10 \times 0.292227 = 1.02922 \implies z^2 = (1.02922)^2 \approx 1.05930$$

3. Time of Carburization:

$$x^2 = (6.0 \times 10^{-4}\ \mathrm{m})^2 = 3.60 \times 10^{-7}\ \mathrm{m}^2$$

$$4 \cdot z^2 \cdot D = 4 \times 1.05930 \times (1.28 \times 10^{-11}\ \mathrm{m}^2/\mathrm{s}) = 5.4236 \times 10^{-11}\ \mathrm{m}^2/\mathrm{s}$$

$$t = \frac{3.60 \times 10^{-7}\ \mathrm{m}^2}{5.4236 \times 10^{-11}\ \mathrm{m}^2/\mathrm{s}} \approx \mathbf{6638}\ \mathrm{s} = \mathbf{1.844}\ \mathrm{h}$$

Discrepancy with the official solution. The official key gives $t = 6636$ s. Linear interpolation in the table gives $z = 1.0292$ and $t = 6637.6$ s ≈ 6638 s (the exact inverse error function, $z = 1.0271$, would give 6665 s because the table is linear between 1.00 and 1.10). The key is 0.02% below the interpolated value; this is a rounding difference, and the interpolated value is reported.

PHASE 4 · PHYSICAL & ENGINEERING VERIFICATION

Dimensional Check: $[t] = \frac{\mathrm{m}^2}{\mathrm{m}^2/\mathrm{s}} = \mathrm{s}$. **Physical Insight:** About 1.84 hours at $927^\circ\mathrm{C}$ bring the carbon content at $0.6\ \mathrm{mm}$ to $0.30\ \mathrm{wt}\%$. Carbon is small, so it diffuses interstitially [Slide 13]; in the surface layer it blocks the movement of dislocations and hardens the steel for wear resistance, while the core stays tough, and the excess carbon leaves residual compressive stresses [Slide 30].